Notes: Begley and Purnanandam (JFE, 2021)

Color and Credit: Race, Regulation, and the Quality of Financial Services

Contents

Color and Credit: Race, Regulation, and the Quality of Financial Services 1

1 One-sentence summary 2

2 Big picture 2
- 2.1 Main question . . . 2
- 2.2 Core result . . . 2
- 2.3 Regulation mechanism . . . 2
- 2.4 Main policy lesson . . . 2

3 Incremental contribution of the paper 2
- 3.1 Quality dimension of financial discrimination . . . 2
- 3.2 CRA as a quantity shock . . . 2
- 3.3 Race-income interaction . . . 3
- 3.4 New data source . . . 3
- 3.5 Matched local empirical design . . . 3

4 Data and measurement 3
- 4.1 CFPB complaints . . . 3
- 4.2 Demographic variables . . . 3
- 4.3 CRA target definition . . . 3
- 4.4 Complaint ratio and resolution measures . . . 4

5 Empirical specifications 4
- 5.1 Demographic gradient in complaints . . . 4
- 5.2 CRA matching estimator . . . 4
- 5.3 Quantity-quality two-stage model . . . 4

6 Identification and empirical design 4
- 6.1 What is identified . . . 4
- 6.2 Why CRA creates useful variation . . . 4
- 6.3 Why matching is needed . . . 5
- 6.4 Minority heterogeneity as mechanism evidence . . . 5
- 6.5 Main limitations . . . 5

7 Main results 5
- 7.1 Demographic gradients . . . 5
- 7.2 CRA target effects . . . 5

7.3 Race and CRA interaction 5
7.4 Quantity-quality trade-off 5
7.5 Robustness and economic significance 5

8 Tables and figures **6**
8.1 Figure 1: Example geographical variation in low-to-moderate income designation; Table 1: Sample summary statistics 6
8.2 Table 2: Income, education, and race; Figure 2: Minority share and quality 6
8.3 Figure 3: Area income and CRA-target status; Figure 4: Covariate balance for raw and matched samples 7
8.4 Table 3: CRA-target areas and complaints within MSA; Table 4: CRA-target areas and complaints across the country 7
8.5 Table 5: Quantity-quality trade-off, two-stage regressions 8
8.6 Table 6: Alternative measure, complaint ratio; Table 7: Complaints by resolution 9
8.7 Figure 5: Consumer complaints and fines paid by banks; Table 8: Alternative matching criteria 9

9 Short summary **10**

10 Conclusion **10**
10.1 Core conclusion 10
10.2 Why the result matters 10
10.3 Policy implication 10
10.4 Limitations 11
10.5 Takeaway 11

1 One-sentence summary

Begley and Purnanandam show that consumer financial complaints are higher in poorer and minority neighborhoods and that CRA-induced increases in mortgage lending to low-income areas are associated with worse service quality, especially in high-minority areas.

2 Big picture

2.1 Main question

The paper asks whether quantity-focused consumer-credit regulation can unintentionally worsen the quality of financial services. The setting is U.S. mortgage markets, and the quality measure is consumer complaints to the Consumer Financial Protection Bureau (CFPB).

2.2 Core result

Mortgage-related complaints are more common in low-income, less educated, and high-minority zip codes. The minority gradient is particularly steep: moving from low-minority to very-high-minority areas is associated with a large increase in complaints.

2.3 Regulation mechanism

The Community Reinvestment Act (CRA) designates low-to-moderate-income tracts as targets for bank lending. The regulation increases pressure on banks to expand credit access. The paper argues that this positive quantity shock can lower service quality by inducing banks to expand aggressively into harder-to-serve or less sophisticated markets.

2.4 Main policy lesson

Regulation that emphasizes credit quantity but does not directly monitor service quality can create a quantity-quality trade-off. The intended beneficiaries of financial inclusion policy may receive more credit but lower quality financial services.

3 Incremental contribution of the paper

3.1 Quality dimension of financial discrimination

Much previous work studies access to credit or mortgage pricing. This paper studies a different margin: the quality of financial services, measured by complaints about misselling, fraud, poor servicing, and other customer problems.

3.2 CRA as a quantity shock

The paper uses the CRA target designation to study a regulation-driven increase in credit supply. This lets the authors connect financial regulation directly to service quality, not just credit volume.

3.3 Race-income interaction

The paper shows that the adverse quality effect of CRA pressure is concentrated in high-minority areas. Thus, the paper contributes to the literature on unequal financial treatment by showing that minority consumers may face lower service quality even inside programs intended to expand access.

3.4 New data source

The paper uses CFPB complaint data, which provide a direct window into consumer-facing problems in financial services. This is a meaningful expansion beyond loan denial rates, interest rates, and default outcomes.

3.5 Matched local empirical design

The matching design compares CRA-target zip codes with observationally similar non-target zip codes, often within the same MSA. This reduces concerns that differences in complaints simply reflect differences in local economic conditions or consumer complaint culture.

4 Data and measurement

4.1 CFPB complaints

The main dependent variable is the number of mortgage-related complaints filed with the CFPB from 2012 to 2016 at the five-digit zip-code level. The baseline outcome is

$$\ln Complaints_{zip5}.$$

Complaints include problems such as excessive fees, misrepresentation, poor mortgage servicing, debt collection problems, and unsatisfactory resolution.

4.2 Demographic variables

The main zip-code demographic variables are:

- income: average adjusted gross income from IRS data;
- education: share of adults with at least a bachelor's degree;
- minority share: share of residents who are nonwhite;
- mortgage volume: number of mortgages outstanding;
- population and house-price growth controls.

All continuous independent variables are standardized, so coefficients are interpretable as effects of a one-standard-deviation change.

4.3 CRA target definition

A census tract is CRA low-to-moderate income if the tract's median family income is below 80% of its MSA median. The paper aggregates tract-level CRA status to the zip-code level. A zip code is classified as CRA-target if the majority of its population lives in CRA-target tracts. Control zip codes have zero population in CRA-target tracts.

4.4 Complaint ratio and resolution measures

The paper also considers complaint ratios, complaints by resolution status, and fines paid by banks in CFPB enforcement actions. These robustness checks ask whether the main results reflect real service problems rather than merely a higher propensity to complain.

5 Empirical specifications

5.1 Demographic gradient in complaints

The first specification links complaints to income, education, and race:

$$\ln Complaints_{zip5} = \rho(IER_{zip5}) + \sum_{b=2}^{50}(Mort_{b,zip5} + Pop_{b,zip5}) + \zeta_{MSA} + \nu_{zip5}.$$

Here IER is income, education, or race. Mortgage and population bucket controls flexibly absorb scale differences across zip codes.

5.2 CRA matching estimator

The main matching design estimates the average treatment effect on the treated:

$$ATE_T = E[\ln Complaints(1) - \ln Complaints(0) \mid CRA\text{-}target = 1].$$

The propensity score is estimated using mortgage volume, population, income, education, house-price changes, and state indicators. Some specifications restrict matches to the same MSA or to the same income strata.

5.3 Quantity-quality two-stage model

The paper then directly links the CRA designation to lending quantity and quality:

$$quantity_{zip5} = \phi(CRA\text{-}target_{zip5}) + \Gamma X_{zip5} + \zeta_{MSA} + \nu_{zip5},$$

$$\ln Complaints_{zip5} = \psi(\widehat{quantity}_{zip5}) + \Theta X_{zip5} + \zeta_{MSA} + \epsilon_{zip5}.$$

The two quantity measures are mortgage applications and mortgage originations.

6 Identification and empirical design

6.1 What is identified

The paper identifies whether areas targeted by a quantity-focused credit regulation experience worse financial-service quality, relative to observationally similar areas not targeted by the regulation.

6.2 Why CRA creates useful variation

CRA designation depends mechanically on relative tract income. Banks face regulatory pressure to increase lending in CRA-target areas. This creates an exogenous-looking supply incentive to increase the quantity of lending in those areas.

6.3 Why matching is needed

CRA-target areas are poorer by construction. The paper therefore matches target areas to non-target areas that are similar in income, education, population, mortgage volume, and house-price changes. Matching within MSA absorbs local regulatory, economic, and cultural factors that could affect complaint rates.

6.4 Minority heterogeneity as mechanism evidence

If quality problems reflect a regulation-induced quantity shock, the effect should be especially strong where CRA pressure overlaps with fair-lending pressure toward minority borrowers. The paper finds exactly this: the CRA-target/control gap is concentrated in high-minority areas.

6.5 Main limitations

The paper cannot observe every low-quality interaction, only complaints. Complaint propensity may vary across groups. The CRA target status is measured at the tract level but complaints are observed at the zip-code level. The design is therefore not a sharp regression discontinuity. The paper addresses these concerns with matching, complaint-ratio measures, resolution-type tests, and enforcement-fine evidence.

7 Main results

7.1 Demographic gradients

Minority share is strongly associated with complaints, even after controls. The effect of minority share is larger than the effect of income or education, and the relationship is convex: the highest-minority areas have the largest complaint increases.

7.2 CRA target effects

CRA-target areas have more complaints than matched non-target areas. The within-MSA estimate is about 28% more complaints, while broader across-country matching produces larger effects.

7.3 Race and CRA interaction

The target-control complaint gap is statistically indistinguishable from zero in low-minority areas but large in high-minority areas. This suggests that regulation-induced quantity pressure is especially costly where low-income targeting and minority-lending pressure overlap.

7.4 Quantity-quality trade-off

The two-stage regressions show that CRA-target status increases lending quantity and that increases in quantity are associated with higher complaints. This supports the paper's central interpretation: increasing the quantity of credit can reduce service quality.

7.5 Robustness and economic significance

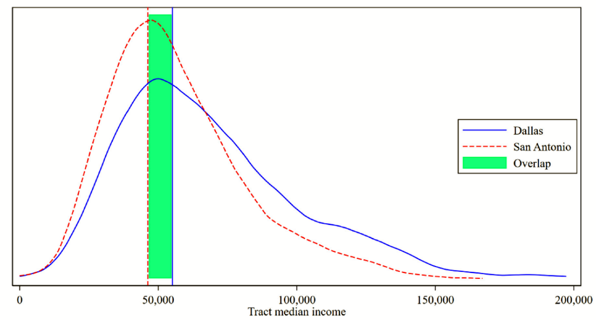
Results are robust to complaint ratios, complaint resolution categories, alternative matching procedures, and multiple bandwidths/kernels. The positive relation between complaints and CFPB fines suggests complaints are economically meaningful.

8 Tables and figures

8.1 Figure 1: Example geographical variation in low-to-moderate income designation; Table 1: Sample summary statistics

Read: Figure 1 illustrates why CRA designation is relative to MSA income: a tract with the same absolute income can be LMI in one MSA but not another. Table 1 summarizes the zip-code-level variables.

Economic message: The figure motivates local comparisons. The table shows the scale and heterogeneity of the complaint data, income, minority shares, mortgage volume, and CRA exposure.



g. 1. Example geographical variation in low-to-moderate income designation.

Figure 1: LMI designation depends on MSA-relative income thresholds.

Table 1
Sample summary statistics.
This table presents the sample summary statistics. *Complaints* is the number of mortgage-related complaints filed to the CFPB in a given five-digit zip code in a given year winsorized at the 1% tails, *lnComplaints* is the log of complaints, *Income* is the mean household adjusted gross income in the five-digit zip code for 2012, *lnIncome* is the log of adjusted gross income, and *College education* is the portion of the adult population in the five-digit zip code with at least a bachelor's degree in 2012. *Minority* is the share of the zip5 population that is nonwhite for 2012. *CRA-target* is an indicator variable equal to one for zip codes with a majority of its population in low- to moderate-income tracts in 2010, zero if none of the population is in an LMI tract, and dropped for zip codes with a positive but less-than-majority of the population in LMI tracts. *Mortgages* is the number of mortgages in the five-digit zip code in 2012 measured by IRS filings with reported mortgage interest, *Population* is the zip code population in 2010, and %ΔHP is the percentage point change in zip5 house price growth (county house price growth is used for observations with no zip code level house price data). All variables are winsorized at the 1% level.

variable	mean	sd	min	p25	p50	p75	max	N
Complaints	10.33	13.25	1.00	2.00	5.00	13.00	71.00	16,309
lnComplaints	1.63	1.22	0.00	0.69	1.61	2.56	4.26	16,309
Income (000)	64.06	52.97	18.65	42.05	51.23	67.61	1,464.53	16,309
lnIncome	10.93	0.44	10.12	10.65	10.84	11.12	12.54	16,309
College education	0.27	0.16	0.05	0.15	0.22	0.35	0.76	16,309
Minority	0.21	0.21	0.01	0.05	0.13	0.30	0.90	16,309
CRA-target	0.28	0.45	0.00	0.00	0.00	0.00	1.00	10,974
Mortgages (000)	1.97	2.07	0.04	0.41	1.19	2.92	9.54	16,309
Population (000)	17.20	15.18	0.62	4.78	12.66	26.11	67.05	16,309
$\%\Delta HP_{2007-2012}$	17.73	15.07	58.3	26.75	15.50	6.35	8.99	15,867

Table 1: Zip-code sample summary statistics for complaints, income, education, minority share, CRA status, mortgages, population, and house-price growth.

The first pair links the institutional CRA treatment rule to the empirical variables used throughout the paper.

8.2 Table 2: Income, education, and race; Figure 2: Minority share and quality

Read: Table 2 shows that income, education, and minority share each predict complaints, but minority share is the strongest predictor in the full model. Figure 2 translates minority-share coefficients into percentage complaint increases.

Economic message: The evidence shows that service-quality problems are not only an income phenomenon. Race composition has an independent and larger association with complaint rates.

Table 2
Income, education, and race.
This table presents OLS estimates from the regression of complaints (*lnComplaints*) for a given five-digit zip code (zip5) on measures of income, education, race, and various sets of fixed effects. *lnComplaints* is the log number of mortgage-related complaints filed to the CFPB in a given zip5 during the sample period (2012–2016), *lnIncome* is the log of the average adjusted gross income of households in each zip5 for 2012, *CollegeEducated* is the share of the zip5 adult population for 2012 with at least a bachelor's degree, and *Minority* is the share of the zip5 population that is a minority race for 2012. *MortBucket50* represents a set of dummy variables for 50 equally populated buckets of the number of mortgages outstanding in the zip code (e.g., 700–750 mortgages) measured by IRS filings with reported mortgage interest for 2012. Similarly, *PopBucket50* represents dummy variable for 50 zip code population buckets for 2012. All continuous independent variables are standardized to have a mean of zero and unit variance. Standard errors are clustered by MSA.

	(1)	(2)	(3)	(4)	(5)	(6)
lnIncome			0.11***			0.08**
			(<0.01)			(0.03)
CollegeEducated				0.05**		0.01
				(0.01)		(0.83)
Minority					0.17***	0.16***
					(<0.01)	(<0.01)
MortBucket50 FE	No	Yes	Yes	Yes	Yes	Yes
PopBucket50 FE	No	No	Yes	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16,309	16,309	16,309	16,309	16,309	16,309
R^2	0.42	0.79	0.79	0.79	0.80	0.80

p-values in parentheses

Table 2: OLS regressions of log complaints on income, education, minority share, and controls.

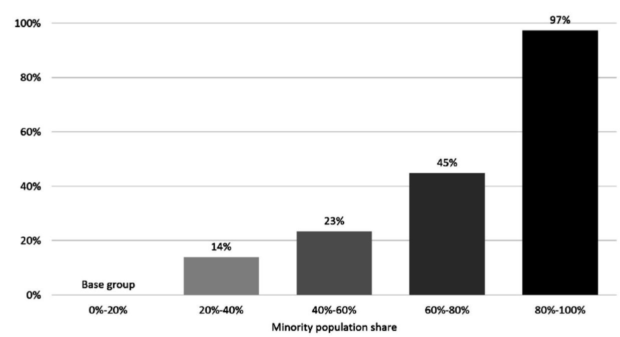


Figure 2: Complaint increases by minority-share bins; the very-high-minority bin has the largest increase.

This pair provides the core demographic-quality fact.

8.3 Figure 3: Area income and CRA-target status; Figure 4: Covariate balance for raw and matched samples

Read: Figure 3 shows that CRA-target zip codes are concentrated in the low-income part of the distribution. Figure 4 shows that matching sharply reduces covariate imbalance between target and control areas.

Economic message: Matching is essential because target and control areas differ mechanically in income. The balance plot shows that the empirical design constructs a more credible comparison group.

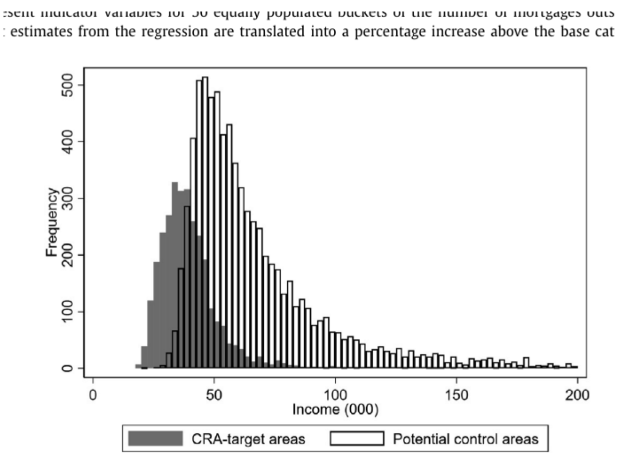


Figure 3: CRA-target areas are concentrated at low income levels relative to potential controls.

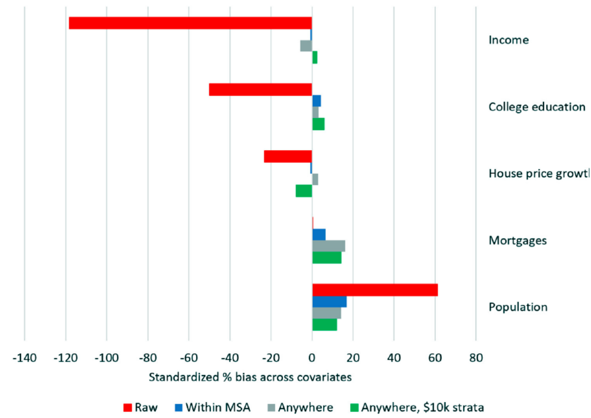


Figure 4: Matching reduces standardized bias across income, education, house-price growth, mortgage volume, and population.

The two figures explain why the paper uses a matched design rather than raw target-control comparisons.

8.4 Table 3: CRA-target areas and complaints within MSA; Table 4: CRA-target areas and complaints across the country

Read: Table 3 shows that CRA-target areas have more complaints than matched controls within the same MSA. The effect is concentrated in high-minority areas. Table 4 confirms the result using a national matching pool.

Economic message: The main effect is not simply local demand or regional complaint culture. The high-minority heterogeneity is key evidence that race and quantity regulation interact.

Table 3

CRA-target areas and complaints: within MSA.

This table presents matching estimates for complaints (*lnComplaints*) for a given five-digit zip code (zip5) for CRA-target zip codes as compared to matched nontarget zip codes, where *CRA-target* is an indicator variable equal to one for zip codes with a majority of its population in low- to moderate-income tracts in 2010. The matching method uses a kernel-weighted propensity score to construct counterfactuals from control zip codes from the same MSA as the treatment zip code to compute the average treatment effect on the treated (ATET). The kernel is epanechnikov with 0.03 bandwidth. The propensity score is estimating using probit regression and includes *lnMortgages, lnPopulation, lnIncome, CollegeEducated*, $\%\Delta HP_{2007-2012}$, and state indicator variables. Column (1) presents the baseline, full sample estimate. Columns (2)–(3), respectively, restrict the sample to zip codes with below- (Low-minority share) and above-median (High-minority share) minority share of the population. *N*treat and *N*control represent the number of matched observations in the treatment and control groups. Standard errors are clustered by MSA.

	(1) Base	(2) Low-minority share	(3) High-minority share
CRA-target	0.25***	0.04	0.35***
	(<0.01)	(0.47)	(<0.01)
N	7082	3431	3430
*N*treat	1701	352	1294
*N*control	5381	3079	2136

p-values in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3: Within-MSA matching estimates; CRA-target areas have more complaints, especially in high-minority areas.

Table 4

CRA-target areas and complaints: across the country.

This table presents matching estimates for complaints (*lnComplaints*) for a given five-digit zip code (zip5) for CRA-target zip codes as compared to matched nontarget zip codes from anywhere in the country. *CRA-target* is an indicator variable equal to one for zip codes with a majority of its population in low- to moderate-income tracts in 2010. The matching method uses a kernel-weighted propensity score to construct counterfactuals to compute the average treatment effect on the treated (ATET). The kernel is epanechnikov with 0.03 bandwidth. The propensity score is estimating using probit regression and includes *lnMortgages, lnPopulation, lnIncome, CollegeEducated*, $\%\Delta HP_{2007-2012}$, indicator variables for each decile of those five variables, and state indicator variables. Columns (1)–(3) use all possible zip codes within the country for matching. Columns (4)–(6) restrict the set of matching zip codes to areas that fall within a $10,000 income strata with treated area. Columns (2)–(3) and Columns (5)–(6) restrict the sample to zip codes with below- (*LowMinority*) and above-median (*HighMinority*) minority share of the population as indicated. *N*treat and *N*control represent the number of matched observations in the treatment and control groups. Standard errors are clustered by MSA.

	All			Income Strata		
	(1) All	(2) LowMinority	(3) HighMinority	(4) All	(5) LowMinority	(6) HighMinority
CRA-target	0.46***	0.09*	0.56***	0.40**	0.12**	0.48*
	(<0.01)	(0.07)	(<0.01)	(0.03)	(0.03)	(0.09)
N	8786	4192	4391	6947	2584	3285
*N*treat	1978	378	1598	1777	343	1333
*N*control	6808	3814	2793	5170	2241	1952

p-values in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Across-country matching estimates; the target-control complaint gap remains positive.

Tables 3 and 4 are the paper's main matching results.

8.5 Table 5: Quantity-quality trade-off, two-stage regressions

Read: Table 5 first shows that CRA target status increases mortgage applications and originations, then shows that predicted loan quantity increases complaints.

Economic message: This table gives the most direct evidence for the paper's quantity-quality trade-off. The regulation increases lending quantity, and the induced quantity is associated with worse service quality.

Table 5

Quantity-quality trade-off: two-stage regressions.

This table presents regression estimates that link the quantity of lending activity to the quality of financial services. The two measures of the quantity of lending activity are the (log) number of applications (*lnApps*) for mortgages and (log) number of mortgages originated (*lnOrigs*) in the zip code during our sample period. Columns (1) and (2) estimate the effect of these measures of the quantity on number of complaints using an OLS regression model. Columns (3) and (4) present the first-stage regression estimates where the instrument is the CRA-target area, which is an indicator variable that equals one if the majority of the population in the zip code are in tracts classified as low-to-moderate income by the CRA and equals zero if the zip code contains no LMI census tracts. Columns (5) provides the reduced-form estimation result linking CRA-target status to the (log)number of complaints (*lnComplaints*). Columns (6)–(7) provide the second-stage estimation for applications and originations, respectively. *Controls* include *lnMortgages, lnPopulation, CollegeEducated*, $\%\Delta HP_{2007-2012}$ the area's income relative to MSA-level median family income. *Controls interactions* contains the interactions of the control variables with the area's relative income. We also include relative income interacted with each of the other control variables. First-stage F-statistics are presented at the bottom of the table. All estimations include MSA fixed effects, and standard errors are clustered at the MSA level.

	OLS		First stage		Reduced form	IV	
	(1) lnComplaints	(2) lnComplaints	(3) lnApps	(4) lnOrigs	(5) lnComplaints	(6) lnComplaints	(7) lnComplaints
CRA-target			0.071***	0.050**	0.134***		
			(<0.01)	(0.02)	(<0.01)		
lnApps	0.172***					1.884***	
	(<0.01)					(<0.01)	
lnOrigs		0.100***					2.666**
		(<0.01)					(0.02)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls interactions	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3126	3126	3126	3126	3126	3126	3126
Instrument						CRA-target	CRA-target
First-stage F-statistic						11.225	5.445

p-values in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Two-stage regressions linking CRA-induced lending quantity to complaints. Since this is a single unpaired table, it is set to width 0.6\linewidth in the LaTeX source.

8.6 Table 6: Alternative measure, complaint ratio; Table 7: Complaints by resolution

Read: Table 6 shows that the result holds when complaints are scaled by mortgage volume. Table 7 shows that results persist across complaint resolution categories, including relief, explanation, and no relief.

Economic message: The main findings are not driven simply by more mortgages or by frivolous complaints. The effect appears in complaint intensity and across complaint-resolution types.

Table 6

Alternative measure: complaint ratio.

This table presents matching estimates for an alternative measure of complaints: number of complaints divided by the number of mortgages in the zip code during the sample period. We standardized this ratio to have mean zero and unit standard deviation. The estimates measure the difference (in standard deviations) in this measure of complaints between CRA-target zip codes and matched nontarget zip codes. The matching method uses a kernel-weighted propensity score to construct counterfactuals to compute the average treatment effect on the treated (ATET). The kernel is epanechnikov with 0.03 bandwidth. The propensity score is estimating using probit regression and includes *lnMortgages, lnPopulation, lnIncome, CollegeEducated*, $\%\Delta HP_{2007-2012}$, indicator variables for each decile of those five variables, and state indicator variables. Column (1) presents the estimate where matches are required to be in the same MSA. Column (2) allows matches from anywhere in the country, and column (3) restricts them to matches to be within the same $10,000 income strata as the treated zip code. *N*treat and *N*control represent the number of matched observations in the treatment and control groups. Standard errors are clustered by MSA.

	Within MSA (1)	Match anywhere (2)	Match within $10k strata (3)
CRA-target	0.40**	0.30***	0.40**
	(0.04)	(<0.01)	(0.03)
N	5765	8526	6757
*N*treat	457	1735	1603
*N*control	5308	6791	5154

p-values in parentheses

Table 6: Complaint ratio estimates.

Table 7

Complaints by resolution.

This table presents matching estimates based on the resolution of complaints. The dependent variable is the standardized (mean zero, unit standard deviation) number of complaints of the relevant type for the zip code. The estimates measure the difference (in standard deviations) in the number of complaints between CRA-target zip codes and matched nontarget zip codes. The matching method uses a kernel-weighted propensity score to construct counterfactuals to compute the average treatment effect on the treated (ATET), and matches are required to be in the same MSA. The kernel is epanechnikov with 0.03 bandwidth. The propensity score is estimating using probit regression and includes *lnMortgages, lnPopulation, lnIncome, CollegeEducated*, $\%\Delta HP_{2007-2012}$, and state indicator variables. Column (1) presents the estimate for the entire sample. Columns (2)-(4) are based on complaints that were resolved with relief, with explanations, and without relief, respectively. *N*treat and *N*control represent the number of matched observations in the treatment and control group. Standard errors are clustered by MSA.

		Resolution of complaint		
	All (1)	Relief (2)	Explanation (3)	No Relief (4)
CRA-target	0.28***	0.19**	0.28***	0.19**
	(<0.01)	(0.04)	(<0.01)	(0.04)
N	5987	5987	5987	5987
*N*treat	606	606	606	606
*N*control	5381	5381	5381	5381

p-values in parentheses

Table 7: Complaints by resolution category.

Tables 6 and 7 address alternative complaint definitions and complaint-quality concerns.

8.7 Figure 5: Consumer complaints and fines paid by banks; Table 8: Alternative matching criteria

Read: Figure 5 shows that banks with more complaints also pay larger CFPB enforcement fines. Table 8 shows that the matching estimates are robust to alternative kernels, bandwidths, and nearest-neighbor specifications.

Economic message: Complaints are economically meaningful enough to predict enforcement fines, and the central target-control result is not tied to one matching procedure.

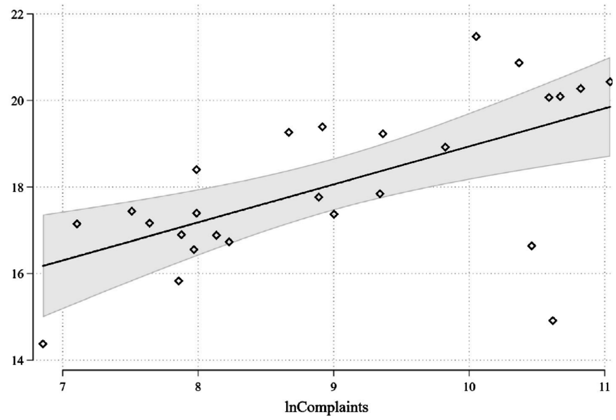


Figure 5: Bank-level complaints are positively related to CFPB fines.

Table 8
Alternative matching criteria.
This table presents matching estimates for alternative matching techniques. The dependent variable is the log number of complaints in the zip code. *CRA-target* is one for zip codes where the majority of the population is in a low-to-moderate income (LMI) census tract as designated by the CRA. Control zip codes are those with zero population in LMI census tracts. The estimates measure the difference in this measure of complaints between CRA-target zip codes and matched nontarget zip codes. Columns (1)-(3) uses kernel-weighted propensity score to construct counterfactuals to compute the average treatment effect on the treated (ATET). The kernel is epanechnikov with bandwidth of 0.03, 0.02, and 0.04 in columns (1), (2) and (3), respectively. Column (4) uses local linear regression matching technique instead of kernel weighting. Columns (5)-(7) are based on nearest-neighborhood matching criteria using one, three, and five nearest neighbors, respectively. The propensity score is estimating using probit regression and includes *lnMortgages*, *lnPopulation*, *lnIncome*, *CollegeEducated*, $\%\Delta HP_{2007-2012}$, and state indicator variables. Each estimate requires matches to be from the same MSA as the CRA-target zip code. *N*treat and *N*control represent the number of matched observations in the treatment and control group. Standard errors are clustered by MSA.

Scheme	(1) k(0.03)	(2) k(0.02)	(3) k(0.04)	(4) llr(0.03)	(5) nn(1)	(6) nn(3)	(7) nn(5)
CRA (atet)	0.25	0.17	0.39	0.43	0.50	0.51	0.39
	(<0.01)	(0.02)	(<0.01)	(<0.01)	(<0.01)	(<0.01)	(<0.01)
N	5924	5874	6004	7064	7064	7064	7064
*N*treat	543	493	623	1683	1683	1683	1683
*N*control	5381	5381	5381	5381	5381	5381	5381

p-values in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Alternative matching methods preserve the main finding.

The final pair supports the economic significance and robustness of the complaint measure.

9 Short summary

- Mortgage complaints are more common in lower-income and high-minority zip codes.
- Minority share is a stronger predictor of complaints than income or education.
- CRA-target areas have more complaints than matched non-target areas.
- The CRA effect is concentrated in high-minority areas.
- Two-stage regressions link CRA-induced increases in lending quantity to higher complaints.
- Results are robust to complaint ratios, resolution types, matching methods, and enforcement-fine evidence.
- The paper documents an unintended adverse quality consequence of quantity-focused credit-access regulation.

10 Conclusion

10.1 Core conclusion

Begley and Purnanandam show that consumer financial-service quality is lower in high-minority and low-income areas, and that CRA-targeted increases in credit supply are associated with more mortgage complaints. The adverse effect is especially strong in high-minority CRA-target areas.

10.2 Why the result matters

The paper shifts attention from the quantity of credit to the quality of credit. Access-oriented regulation can succeed on quantity while worsening consumer experience through more misselling, poor service, servicing problems, or aggressive sales practices.

10.3 Policy implication

Regulators should monitor service quality alongside lending quantity. If CRA evaluations reward lending volumes without equally monitoring consumer outcomes, banks may satisfy access goals while creating new harms for vulnerable borrowers.

10.4 Limitations

Complaints are an imperfect measure of quality because they depend on consumer awareness, willingness to complain, and CFPB access. The CRA treatment is measured at the zip-code level even though designation occurs at the census-tract level. The estimates should therefore be read as evidence of a strong association consistent with a quantity-quality trade-off, rather than a perfect randomized experiment.

10.5 Takeaway

The paper's key message is that financial inclusion policy needs a quality dimension. More credit is not automatically better credit, especially in markets where consumers face complex products and information disadvantages.